

Differential scanning calorimetry (DSC) is a technique used in the biopharmaceutical industry to characterize the thermodynamic properties of large biomolecules. DSC (Figure 1) utilizes an aluminum sample pan filled with a solvent and biomolecule of interest and an identical reference pan filled only with the solvent. Each pan is heated separately at a constant rate while maintaining a temperature difference of zero between the sample and reference.

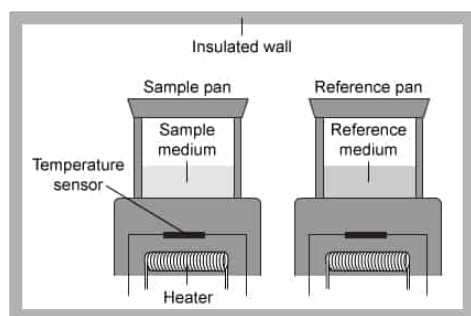


Figure 1 Differential scanning calorimeter

As the temperature of both the sample and reference medium increases, the biomolecule in the sample solution experiences a chemical or physical change, causing a temperature difference relative to the reference medium. To compensate for this temperature difference, the heat flow (heat (q) supplied per unit time (t)) from the instrument heater to the sample pan is increased (or decreased) to return the temperature difference to zero. The difference between the heat q_s needed to raise (or lower) the sample to a specific temperature and the heat q_r needed to raise the reference medium to the same temperature is the excess heat, q_{ex} .

$$q_{ex} = q_s - q_r$$

Equation 1

The excess heat capacity $C_{p,ex}$ of the sample across a temperature range ΔT can be derived from the excess heat flow, the heating rate, and the mass of the biomolecule sample by Equation 2.

$$\left(\frac{q_{ex}}{t}\right)_{flow} = mC_{p,ex}\left(\frac{\Delta T}{t}\right)_{rate}$$

Equation 2

During a DSC experiment investigating the thermodynamics of the reversible and irreversible unfolding of a particular protein, a researcher prepared two buffered sample solutions that each contained 7.5×10^{-3} g of the protein and were purified to remove any nonvolatile solutes.

The protein solutions were each analyzed by DSC and heated at a rate of $1.5^\circ\text{C}/\text{minute}$, resulting in the thermal profiles (thermograms) in Figure 2.

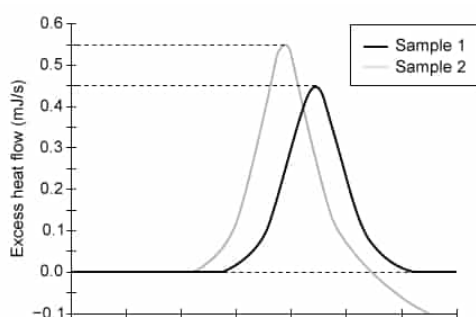
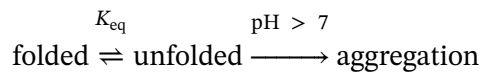


Figure 2 Thermograms of sample 1 (pH = 4.7) and sample 2 (pH = 9.4) after a heating cycle.

The single peak in each thermogram represents the melting temperature (T_m) of the protein. For reversible protein folding, $\Delta G^\circ = 0$ when the temperature reaches T_m .

The unfolded protein strands were observed to refold at pH < 7, but at a pH > 7, the unfolded protein strands were observed to aggregate without refolding (Reaction 1).



Reaction 1

Based on Figure 2, q_s is:

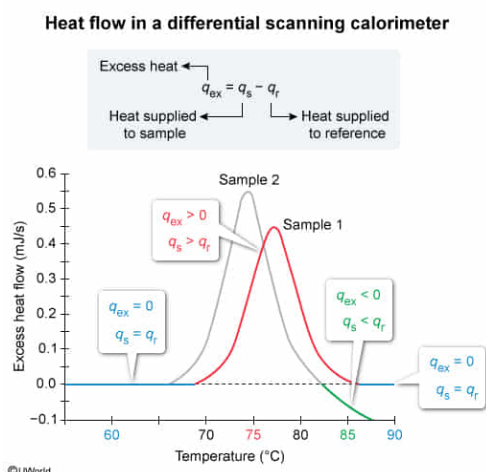
- I. equal to q_r at 60 °C for sample 2.
- II. less than q_r at 75 °C for sample 1.
- III. less than q_r at 85 °C for sample 2.
- IV. equal to zero at 90 °C for sample 1.

- ✓ ☒ A. I and III only [53%]
☐ B. I and IV only [25%]
☐ C. II and III only [12%]
☐ D. II and IV only [8%]

Correct

52% answered correctly

Explanation:



Calorimetry is a technique that measures the amount of heat exchanged between a substance undergoing a chemical or physical change and a calibrated object (ie, a calorimeter). In a differential scanning calorimeter, the amount of **heat absorbed or released** by a sample is measured based on a

temperature difference between the sample and reference material.

For a sample undergoing a thermal transition (eg, a protein unfolding), the amount of heat (or heat flow) needed to increase or decrease the temperature of the sample to the same temperature as the reference medium is referred to as the **excess heat** q_{ex} , which is determined using Equation 1:

$$q_{\text{ex}} = q_{\text{s}} - q_{\text{r}}$$

where q_{s} and q_{r} are the heat supplied to the sample and to the reference medium, respectively.

When interpreting a graph of excess heat flow versus temperature (ie, a thermogram), there are three scenarios:

- $q_{\text{ex}} = 0$: the same amount of heat is needed to raise both the sample and reference to the same temperature ($q_{\text{s}} = q_{\text{r}}$)
- $q_{\text{ex}} > 0$: more heat is needed to raise the temperature of the sample to that of the reference ($q_{\text{s}} > q_{\text{r}}$)
- $q_{\text{ex}} < 0$: heat loss is needed to lower the temperature of the sample to that of the reference ($q_{\text{s}} < q_{\text{r}}$)

According to the thermogram for sample 2, at a temperature of 60 °C (x-axis), $q_{\text{ex}} = 0$ (y-axis). Therefore, $q_{\text{s}} = q_{\text{r}}$ at 60 °C for sample 2 (**Number I**). As the temperature increases to 85 °C, $q_{\text{ex}} < 0$. Therefore $q_{\text{s}} < q_{\text{r}}$ at 85 °C for sample 2 (**Number III**).

(**Number II**) For sample 1, at 75 °C, $q_{\text{ex}} > 0$. Therefore, $q_{\text{s}} > q_{\text{r}}$.

(**Number IV**) For sample 1, at 90 °C, $q_{\text{ex}} = 0$. Therefore, $q_{\text{s}} = q_{\text{r}}$.

Educational objective:

Calorimetry is a technique that measures the amount of heat transferred between a sample and a calorimeter.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403774

System:
Thermodynamics, Kinetics & Gas Laws

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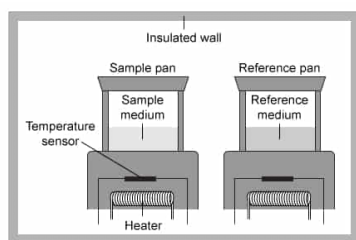


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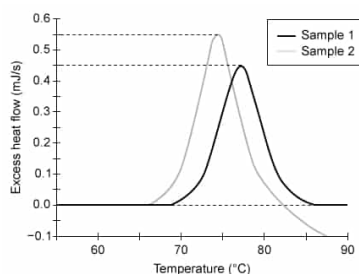
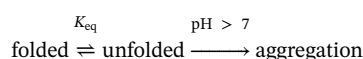


Figure 2 Thermograms of sample 1 (pH = 4.7) and sample 2 (pH = 9.4) after a heating cycle.

The single peak in each thermogram represents the melting temperature (T_m) of the protein. For reversible protein folding, $\Delta G^\circ = 0$ when the temperature reaches T_m .

The unfolded protein strands were observed to refold at pH < 7, but at a pH > 7, the unfolded protein strands were observed to aggregate without refolding (Reaction 1).



Reaction 1

Which statement regarding Reaction 1 is best supported by the data shown in Figure 2?

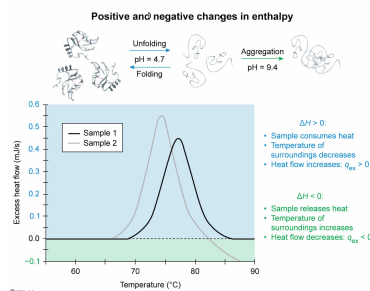
- ☐ A. Both protein unfolding and aggregation result in a negative ΔH . [10%]

- ☐ B. Both protein unfolding and aggregation result in a positive ΔH . [12%]
- ☐ C. Protein unfolding results in a negative ΔH , whereas protein aggregation results in a positive ΔH . [31%]
- ✓ ☒ D. Protein unfolding results in a positive ΔH , whereas protein aggregation results in a negative ΔH . [45%]

Correct

46% answered correctly

Explanation:



When the exchange occurs under constant pressure, the amount of total heat released or consumed by a chemical reaction is equal to the **change in enthalpy** (ΔH) of the reaction.

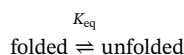
$$\text{Total heat released or consumed} = \Delta H (\text{at constant pressure})$$

Exothermic reactions **release heat** to the surroundings (ie, surrounding temperature increases), resulting in a **negative ΔH** , whereas **endothermic** reactions **consume heat** from the surroundings (ie, surrounding temperature decreases), resulting in a **positive ΔH** .

In a differential scanning calorimeter, the amount of heat absorbed or consumed by a sample is measured based on a temperature difference between the sample and reference material.

- During an **endothermic process**, the **sample temperature decreases** relative to that of the reference. As a result, the sample **needs more heat** added (ie, $q_{\text{ex}} > 0$) than the reference to maintain equal temperatures.
- During an **exothermic process**, the sample temperature **increases** relative to that of the reference. Consequently, the sample **needs less heat** added (ie, $q_{\text{ex}} < 0$) than the reference to maintain equal temperatures.

According to the passage, at a pH < 7, protein unfolding is reversible:



However, at a pH > 7, protein unfolding is irreversible, and the unfolded protein strands aggregate:



Figure 2 shows that with increasing temperature, protein unfolding produces a positive peak ($q_{\text{ex}} > 0$) in the thermograms for both sample 1 (pH = 4.7) and sample 2 (pH = 9.4). This indicates that protein unfolding consumes heat (ie, a **positive ΔH**), which causes the temperature of the sample to decrease. In contrast, sample 2 displays a negative component ($q_{\text{ex}} < 0$) in its thermogram that corresponds to protein aggregation. This indicates that aggregation of unfolded protein strands releases heat (ie, a **negative ΔH**), which causes the temperature of the sample to increase.

(Choices A, B, and C) Protein unfolding *consumes* heat (ie, a $+q_{\text{ex}}$ corresponding to a $+\Delta H$), whereas protein aggregation *releases* heat (ie, a $-q_{\text{ex}}$ corresponding to a $-\Delta H$).

Educational objective:

Exothermic reactions release heat into the surroundings, resulting in a negative ΔH . Endothermic reactions consume heat from the surroundings, resulting in a positive ΔH .

Time spent: 0 sec
Subject:
General Chemistry

QID: 403775
System:
Thermodynamics, Kinetics & Gas Laws

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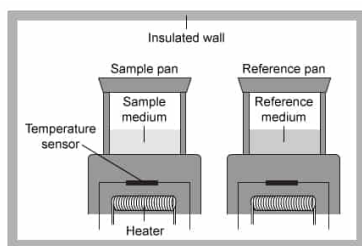


Figure 1 Differential scanning calorimeter

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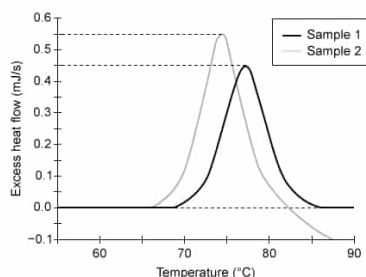
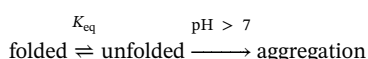


Figure 2 Thermograms of sample 1 (pH = 4.7) and sample 2 (pH = 9.4) after a heating cycle.

The single peak in each thermogram represents the melting temperature (T_m) of the protein. For reversible protein folding, $\Delta G^\circ = 0$ when the temperature reaches T_m .

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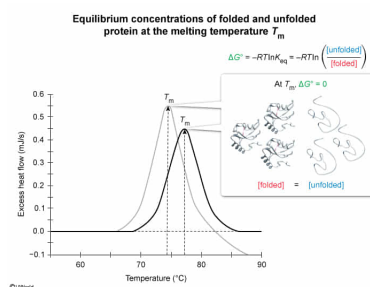
Reaction 1

Which of the following expressions describes the equilibrium concentrations of folded and unfolded protein at T_m for sample 1?

- ☐ A. $\frac{[\text{folded}]}{[\text{unfolded}]} > 1$ [3%]
☐ B. $\frac{[\text{folded}]}{[\text{unfolded}]} = 0$ [8%]
☒ C. $\frac{[\text{unfolded}]}{[\text{folded}]} = 1$ [69%]
☐ D. $\frac{[\text{unfolded}]}{[\text{folded}]} < 1$ [18%]

Correct
68% answered correctly

Explanation:



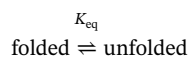
The **equilibrium constant** K_{eq} is the ratio of the equilibrium **molar concentrations** of products to reactants, each raised to the power of their **stoichiometric coefficient**.

$$K_{eq} = \frac{[\text{Products}]^{\text{coefficients}}}{[\text{Reactants}]^{\text{coefficients}}}$$

K_{eq} is related to the standard Gibbs free energy ΔG° of the reaction by the relationship:

$$\Delta G^\circ = -RT \ln K_{eq}$$

where R is the **gas constant** and T is the temperature in **Kelvin**. In sample 1 in the passage, the folded and unfolded protein exist in equilibrium (Reaction 1):



Substituting the concentration ratio corresponding to protein unfolding for K_{eq} into the equation for ΔG° relates the concentration of the folded (reactant) and unfolded (product) species to ΔG° :

$$\Delta G^\circ = -RT \ln \left(\frac{[\text{unfolded}]}{[\text{folded}]} \right)$$

According to the passage, at the melting temperature (T_m) of the protein in sample 1, $\Delta G^\circ = 0$. Therefore, setting $\Delta G^\circ = 0$ and solving for the concentration ratio yields:

$$\begin{aligned}
 0 &= -RT \ln \left(\frac{[\text{unfolded}]}{[\text{folded}]} \right) \\
 e^0 &= e^{\ln \left(\frac{[\text{unfolded}]}{[\text{folded}]} \right)} \\
 1 &= \frac{[\text{unfolded}]}{[\text{folded}]} \Rightarrow [\text{unfolded}] = [\text{folded}]
 \end{aligned}$$

(Choices A and B) The equilibrium concentration ratio for K_{eq} is equal to products *divided* by reactants, not products *multiplied* by reactants. These expressions incorrectly suggest $\left[\text{unfolded} \right] > \frac{1}{[\text{folded}]}$ or $[\text{unfolded}] = 0$, rather than $[\text{unfolded}] = [\text{folded}]$.

(Choice D) A concentration ratio of $\frac{[\text{unfolded}]}{[\text{folded}]}$ 1 would yield $\Delta G^\circ > 0$, *not* $\Delta G^\circ = 0$.

Educational objective:

The equilibrium constant K_{eq} for a reaction is expressed as the ratio of the equilibrium molar concentrations of products to reactant, each raised to the power of their stoichiometric coefficient. The K_{eq} and the standard Gibbs free energy ΔG° of a reaction are related by the equation $\Delta G^\circ = -RT \ln K_{\text{eq}}$.

Time spent: 0 sec
Subject:
General Chemistry

QID: 403776
System:
Thermodynamics, Kinetics & Gas Laws

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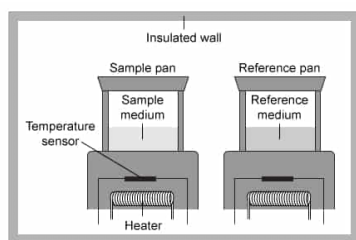


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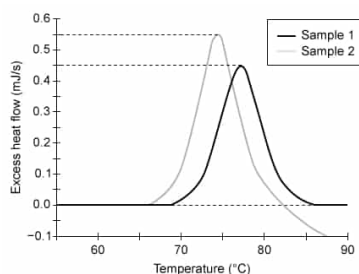
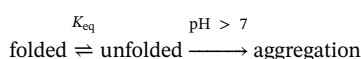


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Reaction 1

What is the maximum $C_{p,ex}$ of sample 1?

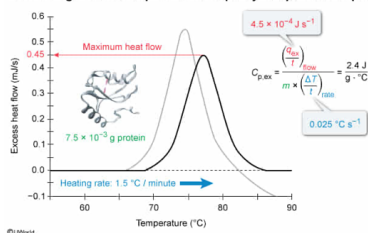
- ☐ A. $0.018 \text{ J}/(\text{g}\cdot^\circ\text{C})$ [12%]

- ☐ B. 0.040 J/(g·°C) [33%]
☒ C. 2.4 J/(g·°C) [40%]
☐ D. 40 J/(g·°C) [13%]

Correct
40% answered correctly

Explanation:

Calculating the excess specific heat capacity of a protein sample



The **specific heat capacity** C_p of a substance (commonly reported in units of J/g·°C) is defined as the amount of heat (q) required to raise the temperature of **1 gram** of a substance by **1 °C**. In other words, the specific heat capacity describes the extent to which 1 gram of a substance will change temperature upon exposure to a given amount of heat.

According to the passage, DSC measures the amount of **excess heat** q_{ex} required to raise the sample temperature to the temperature of the reference at a given point in time. This excess heat is related to the **excess** specific heat capacity $C_{p,\text{ex}}$ by Equation 2:

$$\left(\frac{q_{\text{ex}}}{t}\right)_{\text{flow}} = m C_{p,\text{ex}} \left(\frac{\Delta T}{t}\right)_{\text{rate}}$$

where $(q_{\text{ex}}/t)_{\text{flow}}$ is the excess heat flow, m is the mass of the sample, and $(\Delta T/t)_{\text{rate}}$ is the rate at which the sample is heated. Rearranging the expression to solve for $C_{p,\text{ex}}$ yields

$$C_{p,\text{ex}} = \frac{\left(\frac{q_{\text{ex}}}{t}\right)_{\text{flow}}}{m \left(\frac{\Delta T}{t}\right)_{\text{rate}}}$$

During the DSC experiment described in the passage, a heating rate of 1.5 °C min^{-1} (0.025 °C s^{-1}) is used to heat a sample containing $7.5 \times 10^{-3}\text{ g}$ of protein. Figure 2 shows that the maximum (peak) heat flow is 0.45 mJ s^{-1} ($4.5 \times 10^{-4}\text{ J s}^{-1}$). Substituting these values into the equation above yields the maximum $C_{p,\text{ex}}$:

$$C_{p,\text{ex}} = \frac{\left(\frac{q_{\text{ex}}}{t}\right)_{\text{flow}}}{m \left(\frac{\Delta T}{t}\right)_{\text{rate}}} = \frac{\frac{4.5 \times 10^{-4}\text{ J}}{\text{s}}}{(7.5 \times 10^{-3}\text{ g}) \left(\frac{0.025\text{ °C}}{\text{s}}\right)} = 2.4 \frac{\text{J}}{\text{g} \cdot \text{°C}}$$

(Choice A) A value of $0.018\text{ J/(g} \cdot \text{°C)}$ (with incorrect units) results from dividing the heat flow *only* by the heating rate, which does not account for the mass of the sample and yields the heat capacity of the sample, not the *specific* heat capacity.

(Choice B) A value of $0.040\text{ J/(g} \cdot \text{°C)}$ (with incorrect units) is obtained if the heating rate is not converted from units of °C min^{-1} to °C s^{-1} .

(Choice D) If heating rate is left in units of °C min^{-1} (rather than °C s^{-1}) and heat flow is left in units of mJ s^{-1} (rather than J s^{-1}), a value of $40\text{ J/(g} \cdot \text{°C)}$ (with incorrect units) is calculated.

Educational objective:

The specific heat capacity C_p of a substance is the amount of heat required to raise the temperature of 1 gram of the substance by 1 °C.

Time spent: 0 sec
Subject:
General Chemistry

QID: 403777
System:
Thermodynamics, Kinetics & Gas Laws

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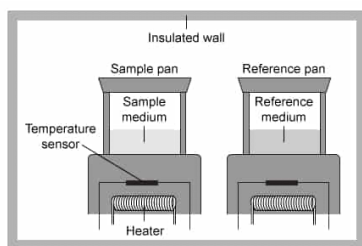


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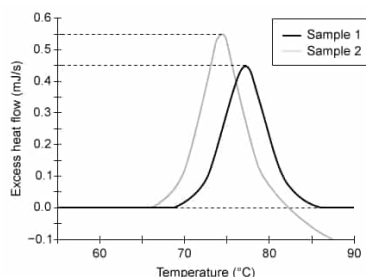
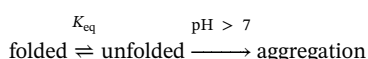


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Reaction 1

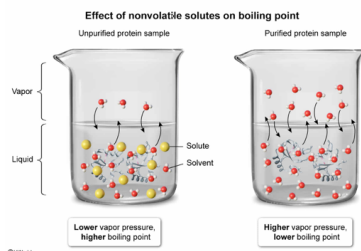
How is the boiling point of the protein solutions affected by the purification described in the passage?

- ☐ A. Increased, because the solution exhibits a lower vapor pressure after the removal of the solutes. [15%]
☐ B. Increased, because the solution exhibits a higher vapor pressure after the removal of the solutes. [22%]
☐ C. Decreased, because the solution exhibits a lower vapor pressure after the removal of the solutes. [34%]
☒ D. Decreased, because the solution exhibits a higher vapor pressure after the removal of the solutes. [27%]

Correct

27% answered correctly

Explanation:



At any given temperature, there is enough energy for some molecules in a liquid to overcome the attractive **intermolecular forces** between the molecules and escape into the gas phase as vapors that are in **equilibrium** with the liquid phase. The pressure resulting from these vapors is called the **vapor pressure**. When the temperature is high enough for the vapor pressure to **equal** the external pressure exerted by the surroundings (ie, ambient pressure), the liquid boils.

Raoult's law states that the vapor pressure of a liquid decreases at any given temperature when a nonvolatile solute (ie, a solute with no vapor pressure) is dissolved in the liquid. This phenomenon occurs because the solute molecules reduce the amount of surface area available for the liquid molecules to occupy and escape into the gas phase. Consequently, the liquid containing the solute must be heated to a **higher temperature** for its vapor pressure to reach the same ambient pressure and begin to boil.

According to the passage, purifying the protein solution removes unwanted nonvolatile solutes. Therefore, purifying the solution results in a **decreased** boiling point due to the purified sample exhibiting a **higher** vapor pressure.

(Choices A, B, and C) Removing dissolved nonvolatile solutes from a liquid *increases* the liquid's vapor pressure, resulting in a *decreased* boiling point.

Educational objective:

The boiling point of a liquid is the temperature at which the vapor pressure of the liquid equals the ambient pressure of the surroundings. According to Raoult's law, the addition of nonvolatile solutes to a liquid decreases the vapor pressure of a liquid, causing the boiling point to increase.

Time spent: 0 sec
 Subject:
 General Chemistry

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 System:
 Thermodynamics, Kinetics & Gas Laws